Homework 1

Due Wednesday, 2 September 2026

1. The opening prices per share Y_1 and Y_2 of two similar stocks are independent random variables, each with density function given by

$$f(y) = \begin{cases} \frac{1}{2}e^{-\frac{1}{2}(y-4)} & y \geq 4 \\ 0 & \text{elsewhere} \end{cases} .$$

   On a given morning an investor is going to buy shares of whichever stock is less expensive.

   a. Find the CDF for the price per share that the investor will pay.
   b. Find the probability density function for the price per share that the investor will pay.

2. Consider a random sample of size n from a distribution with pdf

$$f(x) = \begin{cases} 1/x^2 & 1 \leq x < \infty \\ 0 & \text{else} \end{cases}$$

   a. Find the pdf of the smallest order statistic, $X_{(1)}$.
   b. Find the pdf of the largest order statistic, $X_{(n)}$.
   c. Assuming that n is an odd number, find the pdf of the sample median.

3. In class, we derived the pdf of the largest order statistic for the Uniform(0,1) distribution. Now, suppose that the number of minutes that you need to wait for a bus is uniformly distributed on the interval [0,15].

   a. If you take the bus 5 times, what is the probability that your longest wait is less than 10 minutes? Find the probability theoretically, and also use simulation to estimate the probability.
   b. If you take the bus 5 times, how long should you expect your longest wait to be? Find this value theoretically, and also use simulation to estimate the expected longest wait time.
   c. Rerun your simulations from both (a) and (b) a few (say, 5-10) times. Do the estimated probability and estimated expected longest wait time have the same variability? Comment on what you observe.